

# NAMAN YESHWANTH KUMAR

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## SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) working across ML systems, custom silicon, and compiler-style software. Built SiliconCrew — an automated system driving hardware through synthesis, placement, and verification pipelines, scoring **37/92 on CVDP** and **5/5 on ASU Spec2Tapeout**. Implemented CUDA kernels for neural network training, extended gem5's C++ cache subsystem, and shipped a production ML inference pipeline serving **600+ users**.

## TECHNICAL SKILLS

**Languages & Systems:** C++ | Python | CUDA | SystemVerilog | Verilog | PyTorch | TensorFlow | Async Python | Linux

**ML & Compilers:** LangGraph | Multi-provider ML Inference | Static Timing Analysis | Yosys | OpenROAD | SymbiYosys | gem5 Architectural Simulation

**Infrastructure:** Docker | FastAPI | PostgreSQL | Redis | GitHub Actions CI/CD | DigitalOcean | Nginx

## EDUCATION

**Master of Science — Computer Engineering (GPA 3.81/4.0)**

Arizona State University

May 2026

Tempe, AZ

Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

**Bachelor of Engineering — Electrical and Electronics**

BMS College of Engineering

August 2022

Bengaluru, India

## PROJECT EXPERIENCE

**SiliconCrew — Automated Compiler Pipeline for Hardware Synthesis & Verification**

Nov 2025 – Present

- Built SiliconCrew, an autonomous LangGraph agent (21 tools) that drives a complete EDA pipeline — RTL generation → lint → simulation → Yosys synthesis → OpenROAD place-and-route → post-synthesis simulation → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs, with SymbiYosys formal verification and Cocotb constrained-random; FastAPI + WebSocket backend, MCP server supporting Claude Desktop and VS Code.
- Closes the design loop without human intervention: on simulation failure, reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies autonomously; enforces SDC constraint validation, Yosys RTL-vs-netlist equivalence checking, and signoff guardrails before GDSII tape-out.
- Achieved 37/92 first-pass on CVDP (Comprehensive Verilog Design Problems) and 5/5 on ASU Spec2Tapeout with zero human intervention.

**GCN ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm**

Spring 2025

- Designed an 11-module GCN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved 50,300  $\mu\text{m}^2$  chip area at 50.6% core density, 52.29 mW total power (64.8% switching), and 99.2 ns end-to-end latency.

**Trezzit — Production ML Inference System (trezzit.com)**

Feb 2025 – Present

- Sole-built and shipped a production ML inference system serving **600+ users** for 12+ months — multi-provider pipeline (Gemini, OpenAI, Claude, Grok) improving accuracy from ~60% → **95%+** via two-pass semantic validation with automatic model-upgrade fallback; **995 tests**, GitHub Actions CI/CD, per-provider latency and cost tracking for compute-aware model selection.

**CUDA MLP — Neural Network Training on GPU**

Fall 2024

- Implemented 6 CUDA kernels in C++ for MLP training (linear forward/backward, ReLU forward/backward) — manual GPU memory management (cudaMalloc/cudaMemcpy), kernel launch configuration tuned for occupancy, compiled via nvcc to shared libraries and bound to Python through ctypes FFI; end-to-end training pipeline from raw data to converged model.

**LRU-IPV Cache Replacement Policy — gem5 C++**

Fall 2024

- Implemented LRU with Insertion Position Vector (IPV) replacement policy from scratch inside gem5's C++ source as a first-class SimObject, integrating all 4 cache interface methods (reset, touch, getVictim, invalidate); modified 5 gem5 source files and validated across 3 L2 cache configurations (128/256/512 kB), observing L2 replacements collapse from 637 → 112 → 2 as cache capacity grew — isolating clean capacity-vs-policy effects at 41.8M instructions, 2.04 IPC.

## WORK EXPERIENCE

**Systems Engineer**

NCR GOLD

Oct 2022 – Apr 2024

Bengaluru, India

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

**Research Intern — Propulsion & Power Electronics**

Indian Institute of Science

Oct 2021 – Aug 2022

Bengaluru, India

- Applied Ansys Maxwell FEA to optimize a 1.5 kW axial flux BLDC motor — redesigned stator geometry and upgraded magnetic materials to improve flux linkage and reduce core losses; modeled FOC control and battery management in Simulink for eVTOL propulsion reliability.

## AWARDS & HONORS

- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Won **1st place at Devlabs Hackathon** — HEAL.AI, an AI healthcare assistant with a custom RAG pipeline for insurance Q&A and automated medical bill dispute generation.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.